Example 9.2 (Sums of squares in central factorial numbers). *For integer* $0 \le t \le 5$

$$t = 0: \quad \Sigma^1 n^2 = n \cdot 0 + \tfrac{1}{2}n(1+n) \cdot 0 + \tfrac{1}{6}n(1+3n+2n^2) \cdot 1,$$

$$t = 1: \quad \Sigma^1 n^2 = n \cdot 1 + \tfrac{1}{2}n(n-1) \cdot 2 + \tfrac{1}{6}n(1-3n+2n^2) \cdot 1,$$

$$t = 2: \quad \Sigma^1 n^2 = n \cdot 4 + \tfrac{1}{2}n(n-3) \cdot 4 + \tfrac{1}{6}n(13-9n+2n^2) \cdot 1,$$

$$t = 3: \quad \Sigma^1 n^2 = n \cdot 9 + \tfrac{1}{2}n(n-5) \cdot 6 + \tfrac{1}{6}n(37-15n+2n^2) \cdot 1,$$

$$t = 4: \quad \Sigma^1 n^2 = n \cdot 16 + \tfrac{1}{2}n(n-7) \cdot 8 + \tfrac{1}{6}n(73-21n+2n^2) \cdot 1,$$

$$t = 5: \quad \Sigma^1 n^2 = n \cdot 25 + \tfrac{1}{2}n(n-9) \cdot 10 + \tfrac{1}{6}n(121-27n+2n^2) \cdot 1.$$